

Project Report Draft 1

Introduction

In this report, we aim to explore housing affordability in Metro Vancouver, through analysis of the “Housing Affordability for New Homeowners in B.C., Census Metropolitan Areas Quarterly Data 2025 Q3” dataset obtained from the B.C. Data Catalogue. The complete dataset contains data from all major cities in BC, gathered quarterly. The data was gathered to improve affordability across BC. This data was collected through a combined effort by Statistics Canada, Bank of Canada and BC Assessment, and compiled by BC Stats. They used informational sources such as CRA tax returns, CPI surveys, and property information gathered via BC assessments. Source: <https://catalogue.data.gov.bc.ca/dataset/b-c-housing-affordability/resource/a71e6838-cda1-4acb-adfb-8515c4f9fec9>

As students currently struggling with affordability in Metro Vancouver, we wanted to investigate housing affordability trends in this area and its association to variables such as housing prices, the prime rate, and family income. Specifically, we want to know how interest rates and median family income affect median housing price, after accounting for housing type and number of resales. This will help us understand the key factors driving market price, and potentially derive our own conclusions on affordability trends.

Variables

Variable Name(s)	Type	Description
CMA ID	Integer	Census metropolitan area ID/Name assigned by Statistics Canada, which corresponds to population centres. See the definition on their website.
CMA NAME	Categorical	Census metropolitan area ID/Name assigned by Statistics Canada, which corresponds to population centres.
YEAR	Integer	Each row is specified by a combination of YEAR, QUARTER and HOUSING TYPE. YEAR value ranges from 2000 to 2025.
QUARTER	Integer	QUARTER value ranges from 1 to 4 (Q1–Q4), up to 2025 Q3.
HOUSING TYPE	Categorical	HOUSING TYPE values is one of the values in this set: {Single-detached, Apartment, Row Housing, All Housing Types}.
MEDIAN HOUSING PRICE	Integer	Sale price for a particular year/quarter/house-type. (CAD)
MEDIAN AFTER TAX FAMILY ANNUAL INCOME	Float	Estimated quarterly median family income. (CAD)
PRIME INTEREST RATE	Float	Base loan interest rate used to estimate mortgage rates. (%)
MORTGAGE PAYMENT PERCENT INCOME (MPPI)	Float	Affordability metric based on median after-income tax, home price, minimum down payment, and 25-year variable rate mortgage (based on prime). Higher means less affordable (% of income).
HISTORIC PAYMENT PERCENT INCOME	Float	Long-term running average of MPPI used as benchmark, calculated using MPPI from 2000–2025.
NUMBER OF RESALES	Integer	Resale count per quarter. (Count of homes).

Variable Name(s)	Type	Description
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Table 1: Summary of all variables provided by the dataset, alongside their type, and a short description.

The data was almost entirely clean from the start, there was nothing missing, and no duplicates. Since we're focusing only on Metro Vancouver, the data was filtered to only contain it. The `clean_names` function was also used to make the variable names easier to use for R.

Exploratory Data Analysis

Summary Statistics

Variable	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
<code>cma_id</code>	933	933	933	933	933	933
<code>year</code>	2000	2006	2012	2012	2019	2025
<code>quarter</code>	1.000	1.000	2.000	2.485	3.000	4.000
<code>median_housing_price</code>	150000	345054	481951	580608	683052	1855068
<code>median_after_tax_annual_family_income</code>	42730	53680	64270	67572	79995	99865
<code>prime_interest_rate</code>	2.250	2.969	3.808	4.154	5.258	7.500
<code>mortgage_payment_percent_income</code>	19.92	34.27	42.97	48.96	56.32	141.88
<code>historic_payment_percent_income</code>	33.52	40.56	44.62	49.10	53.17	73.63
<code>number_of_resales</code>	553	2580	4261	5588	7151	22104

Table 2: Summary statistics for all observations in Metro Vancouver CMA.

The dataset contains 412 observations spanning over the years 2000-2025. The median housing price ranges from approximately 150,000 to over 1.85 million, indicating substantial variation in housing values over the observed period. The mean housing price exceeds the median, suggesting a right-skewed plot, prompted by higher-priced properties. To reduce the skewness, stabilize variance, and satisfy the assumptions of linear regression (especially homoscedasticity), **log-transformation** will be applied to `median_housing_price`.

Visualizations

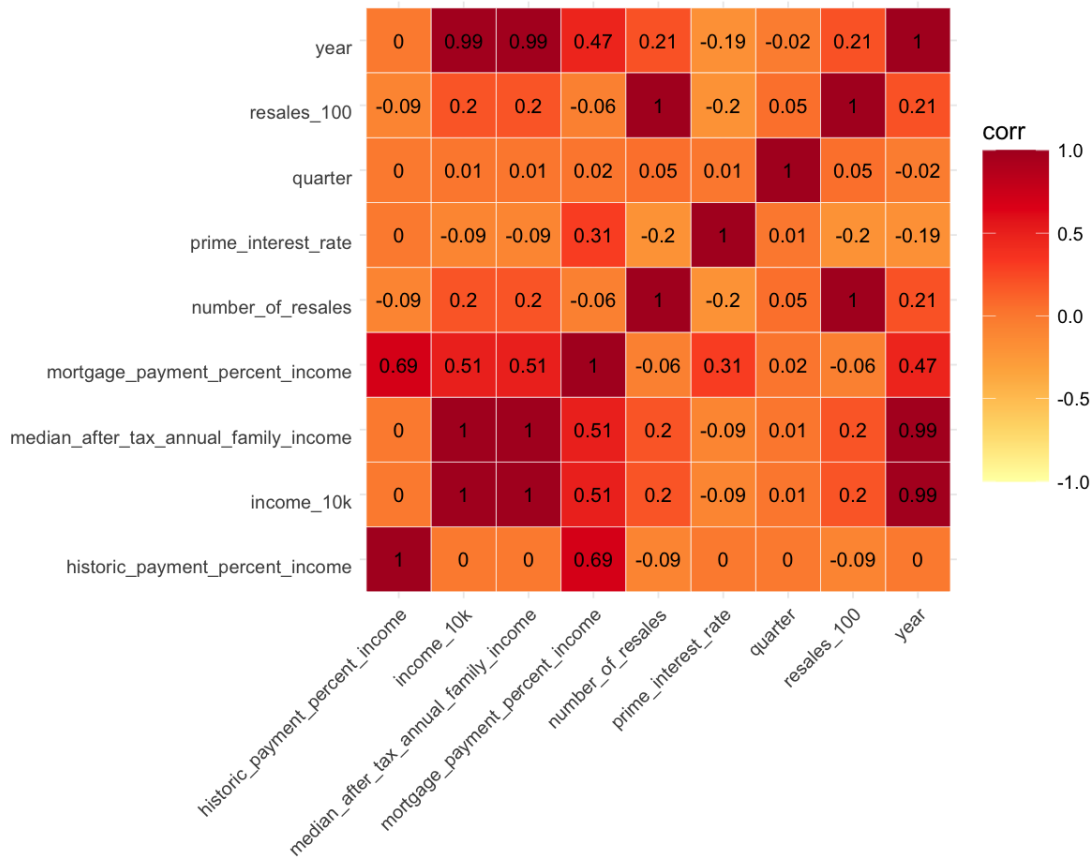


Figure 1: Correlation Coefficient between numeric explanatory variables.

The heatmap above clearly portrays that some of our variables are highly correlated with each other. The covariates 'year' and 'median_after_tax_annual_family_income' seem to have a high correlation coefficient. This indicates potential multicollinearity, suggesting that it may be challenging to isolate the individual effects of these predictors.

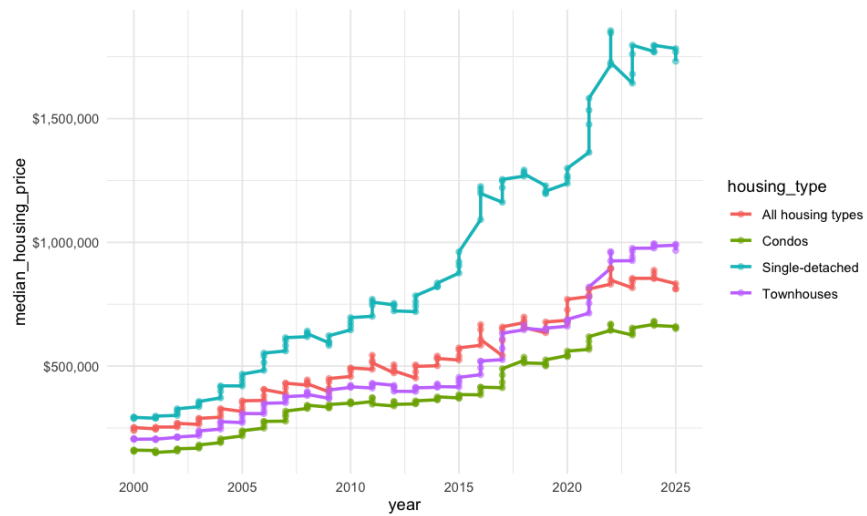


Figure 2: Housing prices over time among different housing types.

The lineplot shows a strong upward trend in median housing prices across all housing types. Single-detached houses consistently have the highest prices, followed by townhouses, while condos remain the least expensive. Price growth also seems to accelerate after 2015.

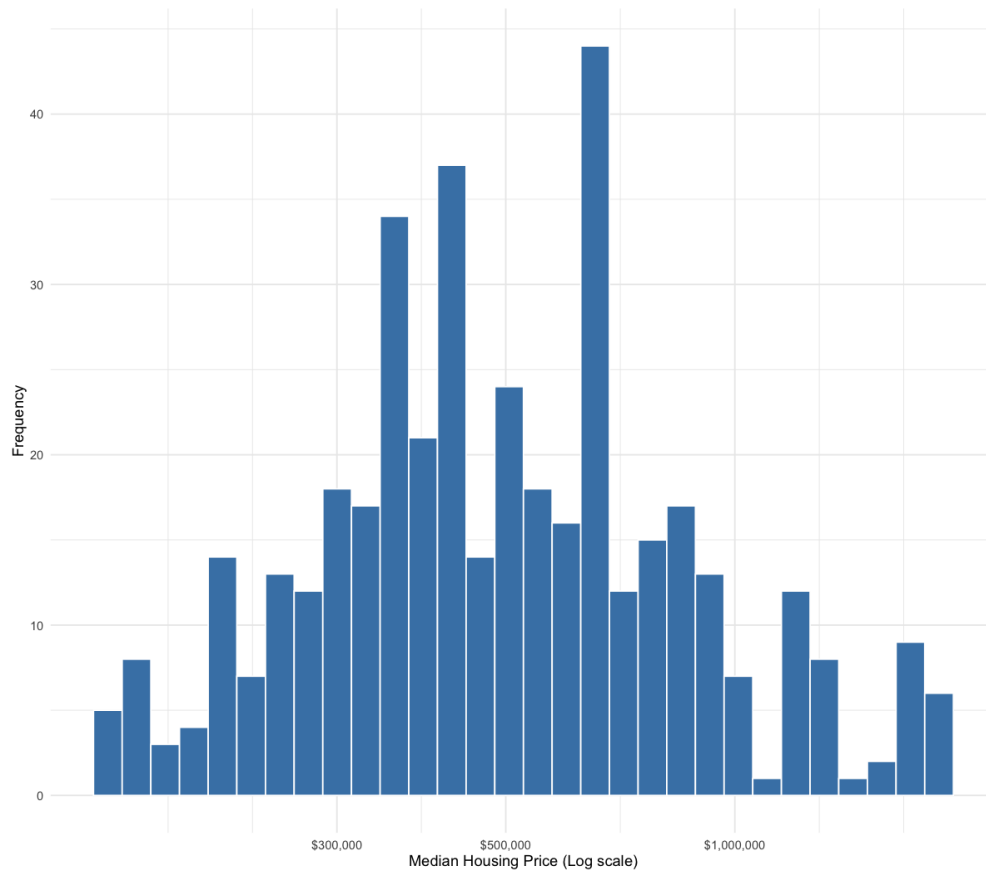


Figure 3: Distribution of log transformed Median Housing Prices

The log-transformed distribution of median housing prices seems approximately symmetric and unimodal, implying that the original right-skewness has been mitigated. The transformation makes the data more suitable for regression analysis.

Statistical analysis

We will scale some of the covariates to make interpreting the result of the model easier, namely dividing `median_after_tax_annual_family_income` by 10000, and `number_of_resales` by 100.

Model Selection

Model :

```
log(median_housing_price) ~ year + quarter + housing_type + prime_interest_rate +
    income_10k + mortgage_payment_percent_income + historic_payment_percent_income +
    resales_100
```

Complete :

	(Intercept)	year	quarter
historic_payment_percent_income	676846/14603		0
	housing_typeCondos	housing_typeSingle-detached	
historic_payment_percent_income	-435524/33949	12877/472	
	housing_typeTownhouses	prime_interest_rate	

```

historic_payment_percent_income -132947/38534      0
                                income_10k    mortgage_payment_percent_income
historic_payment_percent_income      0      0
                                resales_100
historic_payment_percent_income      0

```

From the above, we see that `historic_payment_percent_income` can be expressed as a linear combination of other covariates, so we will remove them in order to find VIF values.

	GVIF	Df	GVIF ^{1/(2*Df)}
year	85.068557	1	9.223262
quarter	1.114402	1	1.055652
housing_type	20.780986	3	1.658101
prime_interest_rate	3.185203	1	1.784714
income_10k	80.281888	1	8.960016
mortgage_payment_percent_income	8.075838	1	2.841802
resales_100	4.851840	1	2.202689

Table 3: GVIF for the predictors in the reduced regression model after removing `historic_payment_percent_income` due to multicollinearity.

From the above, we see that `year`, `income_10k`, `housing_type` all have VIF values >10 , suggesting multicollinearity as we previously pointed out during EDA. We will drop `year` as our research question would like to find relationship between housing prices and other variables including income. From the technical notes in the BC Data Catalogue on our dataset, we know that MPPI (`mortgage_payment_percent_income`) is calculated using median housing prices and prime interest rate, hence we will also drop this variable.

	GVIF	Df	GVIF ^{1/(2*Df)}
housing_type	4.264022	3	1.273415
income_10k	1.156558	1	1.075434
quarter	1.011980	1	1.005972
prime_interest_rate	1.159549	1	1.076823
resales_100	4.602533	1	2.145352

Table 4: GVIF for the predictors in the reduced regression model after removing `historic_payment_percent_income`, `year`, and `mortgage_payment_percent_income` due to multicollinearity concerns.

We observe that after removal of the two covariates, the VIF values have greatly improved to less concerning level below our rule-of-thumb value of 10, suggesting that we have dealt with the multicollinearity problem between our covariates. We will now do model selection to see if we can find a simpler model to interpret with similar performance.

p_params	SSRes	Cp	Adj_R2
6	5.8831	6.83	0.9539540
7	5.8625	7.41	0.9540015
7	5.8671	7.72	0.9539658
8	5.8421	8.00	0.9540480

Table 5: Model comparison results for four candidate subsets using adjusted R^2 and Mallows' C_p .

To finalize our predictive model, we compared four candidate subsets using Adjusted R^2 and Mallows' C_p criteria. Our objective was to select the most parsimonious model, i.e., the simplest version that maintains

high predictive accuracy without introducing bias. As shown in the comparison table, the **Base Model** (including Housing Type, Income, and Prime Rate) achieved an **Adjusted R^2 of 0.9539**. Adding number of resales (**resales**) and seasonality (**quarter**) only increased the R^2 by a negligible **0.0001**, suggesting these variables provide almost no additional explanatory power. Furthermore, the **Mallows' C_p** for the Base Model was **6.83**, which is close to the value we expect (7). This indicates that the model is unbiased and follows the **Principle of Parsimony**. In contrast, the more complex models increased complexity without a meaningful gain in fit. Therefore, the Base Model was selected as the most robust and efficient representation of housing prices in Metro Vancouver.

	GVIF	Df	GVIF ^{1/(2*Df)}
housing_type	1.000000	3	1.000000
income_10k	1.008783	1	1.004382
prime_interest_rate	1.008783	1	1.004382

Table 6: GVIF analysis on final model, showing low likelihood of multicollinearity.

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	11.3	0.0326	348.0	0	11.3	11.4
housing_typeCondos	-0.340	0.0168	-20.3	0	-0.373	-0.307
housing_typeSingle-detached	0.452	0.0168	27.0	0	0.419	0.485
housing_typeTownhouses	-0.104	0.0168	-6.18	0	-0.137	-0.0708
income_10k	0.278	0.00360	77.2	0	0.271	0.285
prime_interest_rate	-0.0283	0.00387	-7.32	0	-0.0359	-0.0207

Table 7: Coefficient estimates and 95% confidence intervals for the final model.

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC
0.955	0.954	0.120	1704.0	7.25e-270	5	291.0	-567.0	-539.0

Table 8: Overall fit statistics for the final regression model.

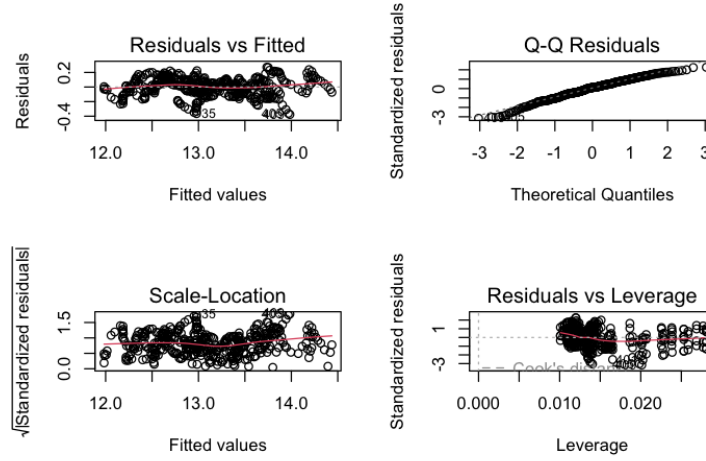


Figure 4: Diagnostic plots for the final linear regression model. The residuals-versus-fitted and scale-location plots suggest roughly constant variance overall, although there is small non-constant spread at the extremes. The Q-Q plot shows a consistent trend, with little deviation even at the extremes, meaning the assumption of normality is unlikely violated.

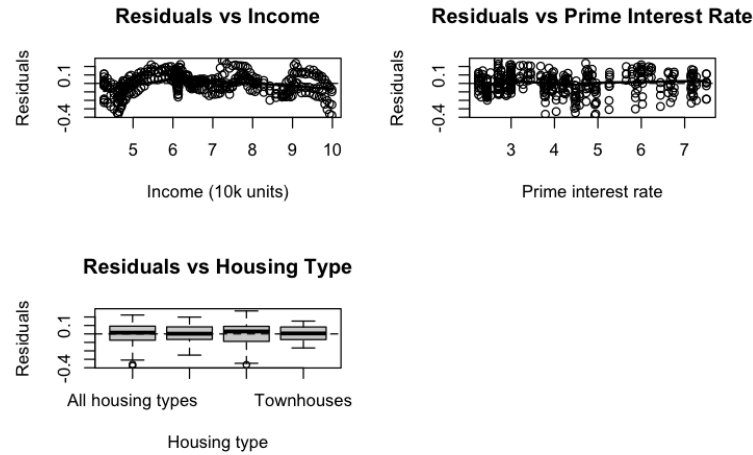


Figure 5: Residuals plotted against covariates in the final linear regression model. The residuals-versus-income plot shows some non-constant variance at the extreme lower and upper ends. The residuals-versus-prime-interest-rate plot shows a more even spread, supporting the constant variance assumption for this predictor. The residuals-housing-type boxplots show a similar spread across categories.

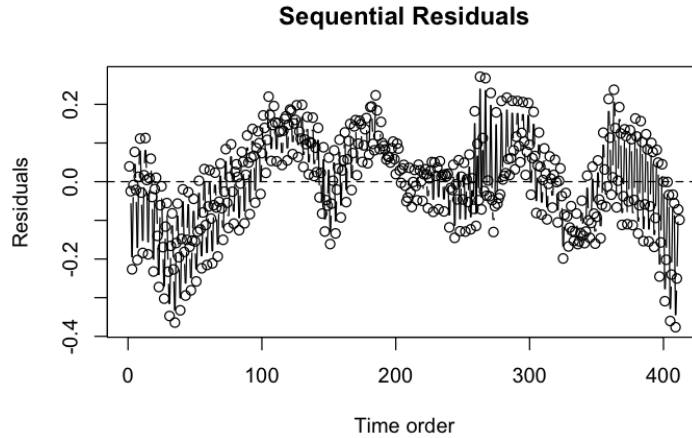


Figure 6: Sequential residual plot for the final linear regression model. The wave-like structure suggests that the errors are not fully independent over time, indicating that the current model may be inadequate.

Discussion

Using linear regression we were able to derive the following model:

$$\log(\text{median_housing_price}) = 11.3 - 0.340 I\{\text{Condo}\} + 0.452 I\{\text{Single_Detached}\} - 0.104 I\{\text{Townhouse}\} + 0.278(\text{income_10k}) - 0.0283(\text{prime_interest_rate})$$

The fitted model observes that a higher median family income is associated with higher median housing prices, whereas higher prime interest rates are associated with lower median housing prices. Furthermore, we notice a difference in price among the different housing types. We expect that single-detached homes have the highest expected prices, followed by townhouses, ‘All housing types’, and then condos. We log-transformed median housing price to help stabilize the variance and make linear regression more appropriate. However, because of that choice, the coefficients above correspond to multiplicative changes in median housing price rather than 1:1 dollar changes. All of the coefficients proved to be significant, with p-values lower than the threshold for a 95% confidence level. Additionally, the 95% confidence interval of our coefficients did not cross zero. Both of these facts can be observed in Table 7. Meaning we can be fairly confident in our results.

Our model is able to explain roughly 95.4% of the variation in log median housing prices. The appropriateness of our model is however up to debate. The plots in figure 4, namely the residual-fitted plot shows a consistent spread and QQ-plot, show a relatively consistent spread supporting the assumption of constant variance. This assumption is further supported by residual-interest rate covariate plot in figure 5. All of these results support the use of linear regression as an appropriate model for this dataset.

On the other hand, the residual-median family income plot showed concerning trends at the extreme left and right ends suggesting heteroscedastic variance. Additionally, the sequential residual plot shows a clear “wave-like” pattern suggesting a lack of independence among sequential observations. These two facts suggest that the errors we found are not fully independent. This makes some sense, as our current model does not directly factor in time. The housing markets are affected by time and economic/political environment. These markets tend to go up and down. Some specific examples we can consider are major economic events such as the 2008 American recession, or the recent downturn in economic performance. There appears to be an almost cyclical up then down cycle of housing prices that a simple linear model cannot account for. Furthermore, we were limited in what covariates we could include in our model due to concerns of multicollinearity across the different variables. Year was removed because of its high collinearity with other predictors such as median

income. If we wanted to include year we would have to remove median income, and vice-versa, if we wanted a stable model. Another issue with our model is that it does not account for resales, this is a conscious choice as adding resales as a parameter in our model did not significantly change its accuracy. However, we then must ask the question if we fully addressed our original research question which asks about resale effects.

In terms of the data itself, the study is observational - meaning we can make at most associative determinations from our model. Additionally, the study uses aggregate quarterly measures meaning variables such as median housing price and median family income are the same across different rows in the same year/quarter. Household-level variation cannot be determined, and figuring out the skew of housing prices within the same quarter seems difficult.

In the future, we would prefer to choose a model more focused on time-series data and the non-linear relations between housing price and our covariates. It would also be helpful to have in this new non-linear model other market related variables un-related to housing. For example, it may be interesting to see how inflation, or recession indicators affect the median housing price.

In summary, the model is useful for capturing some relations between median housing price, family income, and interest rates, but it fails to capture hidden temporal relations and the nature of the data may cause slight violations of constant variance and independence assumptions backing the linear regression model.

Appendix

Statement of AI Use

We used AI to help create the latex document, format it, and transcribe tables from our jupyter notebook to the more formattable latex format before creating our PDF for submission.

Prompt: Here is my jupyter notebook, please transcribe all data frames and tables over to a copy-pasteable latex format exactly as is. Do not do any other work, this is to help avoid redundant work on our end purely for transcription.